Assingment-3.1

TASK-1

List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

- **Testing a computer Power Supply Unit (PSU)**
  The multimeter measures the voltage output of individual pins on the PSU connectors to ensure they are delivering the exact required 12V, 5V, and 3.3V to the motherboard.
- **Finding a short circuit on a smartphone logic board**
  The multimeter is set to continuity mode to trace whether a capacitor or power rail has accidentally connected to the ground plane, which prevents the device from turning on.
- **Checking a laptop battery or charging port**
  The multimeter checks the DC voltage across the charging port contacts or battery pins to verify if power is physically entering the device or if the battery cell voltage has dropped below its functional threshold.

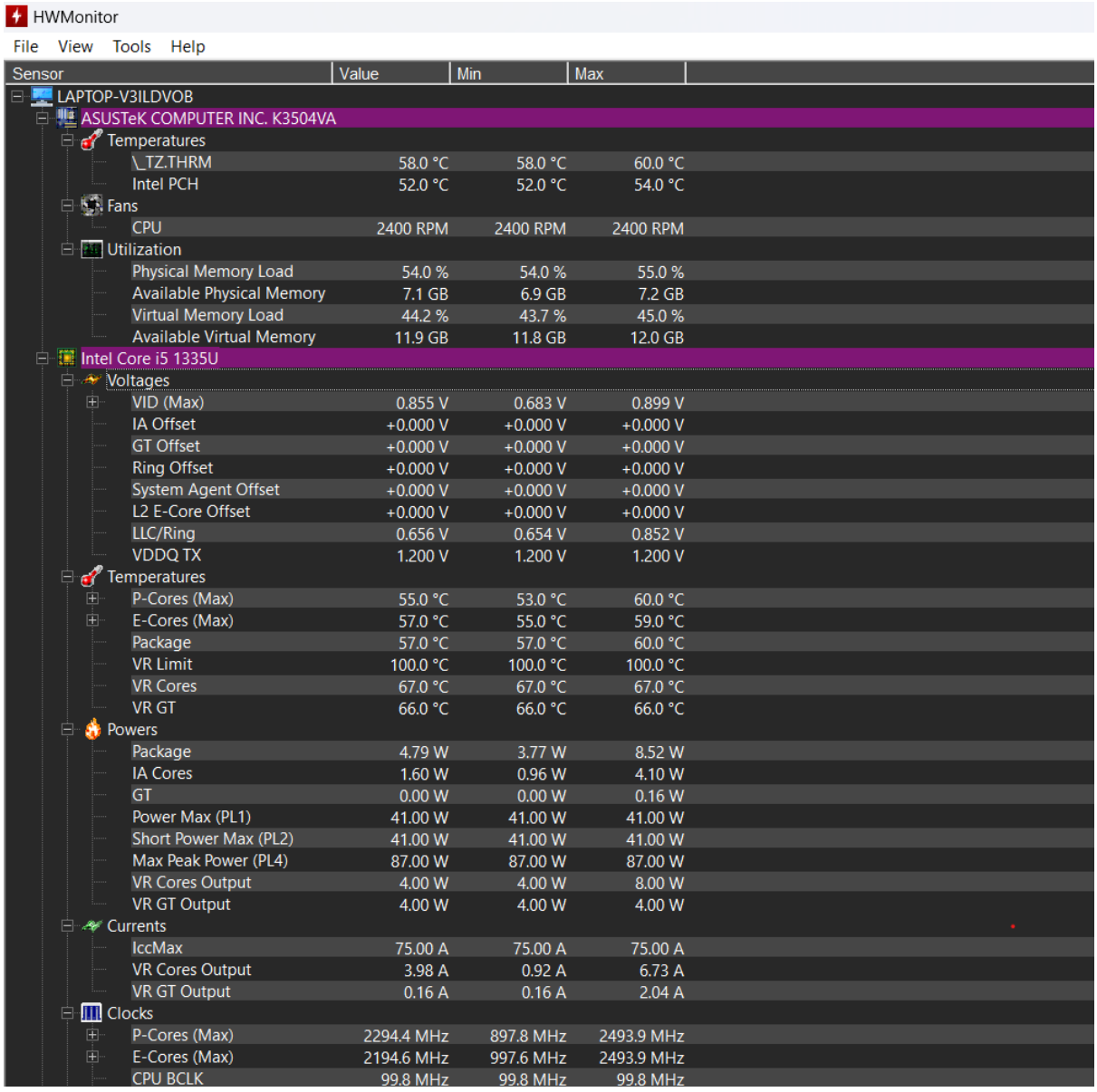
TASK-2

Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

A motherboard POST card features a hexadecimal LED display that cycles through status codes during the boot initialization process. When a system fails to boot, the diagnostic card halts on a specific error code rather than finishing at normal completion codes like FF or 00, helping technicians isolate failures such as unreadable memory modules.

TASK-3

Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.



HWMonitor

File View Tools Help

| Sensor | Value | Min | Max |
|---|---|---|---|
| LAPTOP-V3ILDVOB | | | |
| ASUSTeK COMPUTER INC. K3504VA | | | |
| Temperatures | | | |
| _TZ.THRM | 58.0 °C | 58.0 °C | 60.0 °C |
| Intel PCH | 52.0 °C | 52.0 °C | 54.0 °C |
| Fans | | | |
| CPU | 2400 RPM | 2400 RPM | 2400 RPM |
| Utilization | | | |
| Physical Memory Load | 54.0 % | 54.0 % | 55.0 % |
| Available Physical Memory | 7.1 GB | 6.9 GB | 7.2 GB |
| Virtual Memory Load | 44.2 % | 43.7 % | 45.0 % |
| Available Virtual Memory | 11.9 GB | 11.8 GB | 12.0 GB |
| Intel Core i5 1335U | | | |
| Voltages | | | |
| VID (Max) | 0.855 V | 0.683 V | 0.899 V |
| IA Offset | +0.000 V | +0.000 V | +0.000 V |
| GT Offset | +0.000 V | +0.000 V | +0.000 V |
| Ring Offset | +0.000 V | +0.000 V | +0.000 V |
| System Agent Offset | +0.000 V | +0.000 V | +0.000 V |
| L2 E-Core Offset | +0.000 V | +0.000 V | +0.000 V |
| LLC/Ring | 0.656 V | 0.654 V | 0.852 V |
| VDDQ TX | 1.200 V | 1.200 V | 1.200 V |
| Temperatures | | | |
| P-Cores (Max) | 55.0 °C | 53.0 °C | 60.0 °C |
| E-Cores (Max) | 57.0 °C | 55.0 °C | 59.0 °C |
| Package | 57.0 °C | 57.0 °C | 60.0 °C |
| VR Limit | 100.0 °C | 100.0 °C | 100.0 °C |
| VR Cores | 67.0 °C | 67.0 °C | 67.0 °C |
| VR GT | 66.0 °C | 66.0 °C | 66.0 °C |
| Powers | | | |
| Package | 4.79 W | 3.77 W | 8.52 W |
| IA Cores | 1.60 W | 0.96 W | 4.10 W |
| GT | 0.00 W | 0.00 W | 0.16 W |
| Power Max (PL1) | 41.00 W | 41.00 W | 41.00 W |
| Short Power Max (PL2) | 41.00 W | 41.00 W | 41.00 W |
| Max Peak Power (PL4) | 87.00 W | 87.00 W | 87.00 W |
| VR Cores Output | 4.00 W | 4.00 W | 8.00 W |
| VR GT Output | 4.00 W | 4.00 W | 4.00 W |
| Currents | | | |
| IccMax | 75.00 A | 75.00 A | 75.00 A |
| VR Cores Output | 3.98 A | 0.92 A | 6.73 A |
| VR GT Output | 0.16 A | 0.16 A | 2.04 A |
| Clocks | | | |
| P-Cores (Max) | 2294.4 MHz | 897.8 MHz | 2493.9 MHz |
| E-Cores (Max) | 2194.6 MHz | 997.6 MHz | 2493.9 MHz |
| CPU BCLK | 99.8 MHz | 99.8 MHz | 99.8 MHz |

TASK-4

Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

| Tool | What it checks | Example Problem Detected |
|---|---|---|
| Multimeter | Electrical values like voltage, current, and resistance from power components. | A failing Power Supply Unit (PSU) delivering 10.5V instead of the required 12V to the motherboard. |
| POST Card | Motherboard status codes during the Power-On Self-Test before video initializes | A system hang due to faulty or improperly seated RAM, indicated by a specific hexadecimal code (like C1 or 55). |
| Software Diagnostics | Internal system components, resource stability, and logic errors while the OS or a bootable environment is running. | Severe hard drive sector corruption or read/write failures detected during a SMART scan. |